



Properties of Relations

1. Reflexive

If ordered pair (a, a) is an element of the set of Relation for every a that is an element of set A .
Example:

$$A = \{1, 2, 3\}$$

$$R = \{(1, 1), (2, 2), (2, 3), (3, 3)\} \rightarrow \text{Reflexive}$$

$$R = \{(1, 1), (2, 1), (2, 3), (3, 3)\} \rightarrow \text{Not reflexive}$$

A relation is reflexive if, we observe that for all values $a: a R a$

In other words, all values are related to themselves.

2. Irreflexive

If ordered pair (a, a) is NOT an element of the set of Relation for every a that is an element of set A .
Example: $A = \{1, 2, 3\}$

$$R = \{(1, 3), (2, 1), (2, 3), (3, 1)\} \rightarrow \text{Irreflexive}$$

$$R = \{(1, 1), (2, 2), (3, 3)\} \rightarrow \text{Not irreflexive}$$

3. Symmetric

If ordered pair (a, b) and (b, a) are elements of the set of Relation for every a and b that is an element of set A .

$$\text{Example: } A = \{1, 2, 3\}$$

$$R = \{(1, 1), (2, 1), (1, 2), (3, 3)\} \rightarrow \text{Symmetric}$$

$$R = \{(1, 1), (2, 1), (2, 3), (3, 3)\} \rightarrow \text{Not symmetric}$$

4. Antisymmetric

If ordered pair (a, b) and (b, a) are elements of the set of Relation given that $a = b$.

$$\text{Example: } A = \{1, 2, 3\}$$

$$R = \{(1, 1), (2, 1), (1, 2), (3, 3)\} \rightarrow \text{Not}$$

antisymmetric

$$R = \{(1, 1), (2, 1), (2, 3), (3, 3)\} \rightarrow \text{antisymmetric}$$

Note: Antisymmetric is not the same as "not symmetric".

5. Transitive

If ordered pair (a, b) and (b, a) are elements of the set of Relation, then (a, c) is an element of the set of Relation.

$$\text{Example: } A = \{1, 2, 3\}$$

$$R = \{(1, 1), (2, 1), (1, 2)\} \rightarrow \text{Transitive}$$

$$R = \{(2, 2), (3, 1), (1, 3)\} \rightarrow \text{Not transitive}$$

Equivalence is when relations are reflexive, symmetric and transitive at the same time. $x \sim y$.

A relation satisfies **trichotomy** if we observe that for all values a and b it holds true that $a R b$ or $b R a$.

The relation **is-greater-or-equal** satisfies since, given 2 real numbers a and b , it is true that whether $a \geq b$ or $b \geq a$ (both if $a = b$)



Shorter Notations

Let:

R is the relation between A and B

$D(A)$ = diagonal of set A

$D(A) = \{(a, a) \mid a \text{ in } A\}$

R^{-1} = inversion of relation

$R^{-1} = \{(a, b) \mid (b, a) \text{ in } R\}$

R is **transitive** if and only if $R \bullet R$ is a subset of R .

R is **reflexive** if and only if $D(A)$ is a subset of R .

R is **symmetric** if R^{-1} is a subset of R .

R is **antisymmetric** if and only if the intersection of $D(A)$ and R is empty.

R is **asymmetric** if and only if the intersection of $D(A)$ and R is empty.

R is **function** if and only if $R^{-1} \bullet R$ is a subset of $D(B)$.

In this case it is a function $A \rightarrow B$. Let's assume R meets the condition of being a function, then:

R is **injective** if $R \bullet R^{-1}$ is a subset of $D(A)$.

R is **surjective** if $\{b \mid (a, b) \text{ in } R\} = B$

What are Functions?

A **function** is a relationship between two sets of numbers. $f: A \rightarrow B$

Domain – set of inputs

Codomain – set of outputs

Range – set of all actual outputs

Types of Functions

Functions can either be **one to one** (injective), **onto** (surjective) or **bijective**.

Injective Functions

These are functions in which every element in the domain maps into a unique elements in the codomain.

One-to-One Functions

A function f is one-to-one if and only if $f(a) \neq f(b)$ whenever $a \neq b$.

Surjective Functions

These are functions in which every element in the codomain is mapped by an element in the domain.

Onto Functions

For every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

Bijective Functions

These are functions that are both injective and surjective.

One-to-one Correspondence

If the function is both one-to-one and onto.

Inverse Functions

The inverse function of f is denoted by f^{-1} , where $f(a) = b$ and $f^{-1}(b) = a$ provided that f is a one-to-one function.

$$f(f^{-1}(x)) = x$$

Composition of Functions

Function composition is applying one function to the result of the other. The composition of the function f and g , denoted by $f \circ g$, is defined by:

$$(f \circ g)(x) = f(g(x))$$

LOGIC

Proposition

A declarative sentence that is either T or F.

Ex: Speed of light = 300 million m/s

Compound Proposition

At least two propositions connected by logical connectives

Formal Name	Nickname	Symbol
Negation	NOT	$\neg$
Conjunction	AND	$\wedge$
Disjunction	OR	$\vee$
Exclusive-OR	XOR	$\oplus$
Implication	IMPLIES	$\rightarrow$
Biconditional	IFF	$\leftrightarrow$

Negation $\neg$

It is the opposite of the truth value of a proposition. It can be formed by writing "It is not true that..." or "It is false that..."

Ex: $p = \text{"I am brave."} \quad \neg p = \text{"I am not brave."}$

Conjunction $\wedge$

Propositions can be combined by the word AND to form a compound proposition

Ex: $p \wedge q = \text{"I am talking and you are listening."}$

Disjunction $\vee$

Propositions can be combined by the word OR to form a compound proposition.

Ex: $p = \text{"I am talking."} \quad q = \text{"You are listening."}$

$p \vee q = \text{"I am talking or you are listening."}$

Exclusive OR $\oplus$

Only one is true at a time. Both cannot be true or false at the same time.

Ex: $p = \text{"The outcome of the coin toss is a head."}$

$q = \text{"The outcome of the coin toss is a tail."}$

$p \oplus q = \text{"Either the outcome of the coin toss is a head or a tail."}$

" p is true or q is true, but not both"

Implication $\rightarrow$

$p \rightarrow q$ states that p implies q .

p (antecedent) while q (consequent)

Ex: $p = \text{"Loving you is wrong."}$

$q = \text{"I don't want to be right."}$

$p \rightarrow q = \text{"If loving you is wrong, then I don't want to be right."}$

Inverse $\neg p \rightarrow \neg q$

Converse $q \rightarrow p$

Contrapositive $\neg q \rightarrow \neg p$

Biconditional $\leftrightarrow$

States that p is true IF AND ONLY IF q is true.

Ex: $p = \text{"You will top the board exam."}$

$q = \text{"You have obtained the highest score."}$

$p \leftrightarrow q = \text{"You will top the board exam if and only if you have obtained the highest score."}$